Beyond play: cultivating consumer loyalty through compelling flow experiences in video gaming

Mohammad Mahfuzur Rahman
Department of Management, Marketing and Entrepreneurship, Shippensburg University, Shippensburg, Pennsylvania, USA
Jamie Carlson
Department of Newcastle Business School (Marketing), University of Newcastle, Callaghan, Australia
William Powell
Department of Management, Marketing and Entrepreneurship, Shippensburg University, Shippensburg, Pennsylvania, USA, and
Jiehua Chen
Department of Business Administration, Shandong University, Jinan, China

Abstract

Purpose – This study aims to explore how video game design elements contribute to Flow experiences, shaping consumer attitudes and behaviors. By examining the psychological facets of video gaming, the research aims to inform the literature on Flow theory, as well as game development strategies that foster gamer loyalty, purchase intentions and sustained engagement.

Design/methodology/approach – Using Flow as a theoretical foundation, this study examines how specific dimensions of flow including concentration, interactivity, telepresence, collaboration, challenge and goals impact gamers' Flow experiences and subsequent behavioral responses. Due to the "real-world" nature of this research, a self-reported survey was used to collect data from 201 Chinese video gamers via an online questionnaire. Structural equation modeling was used to examine the Flow experience construct, and the relationships between Flow experience, behavioral intentions and game loyalty.

Findings – Concentration, interactivity and telepresence were found to facilitate Flow experience. However, collaboration, challenges and goals were not statistically significant in this context, suggesting that these dimensions may depend on extraneous variables such as the gaming environment or player motivation. Central to this examination, findings also show that Flow experiences positively influence video gaming loyalty and positive behavioral intentions.

Originality/value – This study advances understanding of Flow experiences in video gaming by identifying specific design elements that enhance player engagement and loyalty. These insights extend the application of Flow theory to video gaming and contribute to strategies for cultivating favorable consumer behavior and consumer loyalty in the video gaming industry.

Keywords Video game, Flow experience, Loyalty, Collaboration, Interactivity, China

Paper type Research paper

1. Introduction

The global video game industry has emerged as a dominant force in entertainment, surpassing traditional media such as film and music in revenue and cultural influence. Within this dynamic landscape, the Chinese video game market stands out as the largest entertainment industry globally, projected to reach $137.80bn in revenue by 2025 (Statista, 2025). With 666 million players across various platforms, including consoles, PCs and smartphones (Thomala, 2023), and over 222 new games developed in 2023 alone (GameCompanies, 2023), the market is both vibrant and highly competitive, which shapes consumer gaming experiences in distinctive ways. Given the highly competitive and rapidly evolving nature of the Chinese gaming industry, understanding what drives sustained player engagement becomes essential. One theoretical framework that helps explain psychological mechanisms for deep engagement in interactive experiences is Flow theory.

China's scale and rapid evolution in the gaming industry make it a compelling context for examining Flow experiences. Unlike Western markets, where franchises like *The Legend of Zelda* and *Call of Duty* maintain long life cycles, the average Chinese video game's life expectancy is remarkably short, often limited to one to two years (Lu, 2016). This transience reflects intense market competition and a consumer base that demands continuous novelty and engagement. Flow theory, which

The current issue and full text archive of this journal is available on Emerald Insight at: https://www.emerald.com/insight/0736-3761.htm



Journal of Consumer Marketing
42/6 (2025) 795–809
© Emerald Publishing Limited [ISSN 0736-3761]
[DOI 10.1108/JCM-04-2024-6755]

Received 9 April 2024
Revised 20 September 2024
3 February 2025
8 April 2025
Accepted 18 May 2025

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

emphasizes optimal experiences marked by deep concentration, seamless interactivity and an equilibrium between challenge and skill (Csikszentmihalyi, 1975), may help explain any prolonged video game success in such an environment. However, while prior studies have explored Flow in gaming and addiction (e.g. Huang *et al.*, 2022; Kuo *et al.*, 2016), they often overlook how its specific dimensions – such as collaboration or telepresence – drive consumer outcomes like loyalty in fast-evolving, culturally distinct markets like China. Understanding Flow experience dimensions is vital for uncovering how game design can sustain player engagement. This, in turn, drives critical outcomes like satisfaction, loyalty and repeat purchases in a market where rapid turnover is the norm.

Moreover, cultural and regulatory factors further differentiate the Chinese gaming ecosystem from others. For example, government-imposed playtime restrictions for minors and a dominant preference for mobile gaming over consoles shape the way games are consumed and experienced (Thomala, 2023). Collectivist cultural values also influence gaming behaviors, with collaboration and social interaction playing a more prominent role in China when compared to individualistic Western markets. Examining Flow within this context not only highlights its universal applicability but also uncovers cultural nuances that enrich our understanding of gaming as a global phenomenon. These cultural and regulatory dynamics likely influence how players experience and sustain Flow states, making China a particularly compelling context for investigating the theory's applicability.

This study makes significant contributions to both theory and practice by contextualizing Flow experiences within the socio-cultural and economic landscape of the Chinese video game market. From a theoretical perspective, it extends Flow theory by exploring its multi-dimensional nature and empirically examining how these dimensions influence critical consumer outcomes such as satisfaction, loyalty and repeat purchase intentions. While existing studies have explored aspects of Flow and Flow-like experiences in online gaming and video game addiction (e.g. Huang *et al.*, 2022; Kuo *et al.*, 2016; Mukherjee and Lau-Gesk, 2016; Hew *et al.*, 2023; Jimenez *et al.*, 2019), there is a notable gap regarding the specific constituent dimensions of Flow and their impact on consumer behavior outcomes within the gaming context. This study uniquely positions the concept within the fast-paced, highly competitive Chinese gaming industry, offering new insights into how Flow operates in environments characterized by rapid product turnover and strong regulatory controls. By doing so, it highlights the universal relevance of Flow while addressing the need for more culturally specific investigations, as emphasized by Hew *et al.* (2023).

From a managerial standpoint, the findings of this study offer actionable insights for game developers and marketers aiming to sustain player engagement and extend product life cycles in the Chinese market. By emphasizing the importance of Flow-inducing elements, the study provides a roadmap for designing games that resonate with Chinese consumers. This dual focus on theoretical innovation and practical application equips both scholars and practitioners with a deeper understanding of the factors that drive consumer loyalty and sustained engagement in one of the world's most dynamic gaming markets. Collectively, this study bridges critical gaps in literature and offers a foundation for future research on the interplay between Flow and consumer behavior in diverse cultural and economic contexts.

To address these gaps, our study aims to provide further insight into Flow experiences during gaming consumption and their relationship with consumer behavior outcomes. Specifically, we pose two research questions:

RQ1. What are the constituent dimensions of Flow in the video gaming consumption context?

RQ2. Does Flow influence consumer outcomes, including overall satisfaction with the gaming experience, continued interaction within the gaming environment, and the intention for repeat purchases and future game purchases from the same developer?

By addressing these areas, this study equips developers and marketers with a deeper understanding of how to foster Flow experiences that drive consumer engagement and loyalty, thereby extending the life cycles of games in one of the most challenging and lucrative markets in the world.

The remainder of this article is structured as follows. First, we begin with a review of Flow literature and present our theoretical framework. We then discuss the research design and results. We conclude with implications for consumer marketing theory and practice, research limitations and directions for future research.

2. Literature on flow theory

Flow theory, introduced by Csikszentmihalyi (1975), posits that individuals experience a state of optimal engagement and enjoyment when they are fully immersed in an activity. This is characterized by intense focus, a sense of control and the merging of action and awareness. Flow has been described as an experience so gratifying that people are willing to do it for its own sake, having little concern for what they gain from it, even when it is difficult or dangerous (Csikszentmihalyi, 1990). Flow theory has been explored and validated across diverse domains, shedding light on how immersing oneself in a single activity can lead to the attainment of a Flow state. This theory has been extensively applied to consumer psychology and information systems fields to understand how individuals interact with digital technologies and the factors that contribute to optimal engagement in the domains of video games (e.g. Chen *et al.*, 2018a, 2018b; Cowley *et al.*, 2008; Soutter and Hitchens, 2016; Su *et al.*, 2016).

Flow theory has provided valuable insights into understanding video gaming experiences. Numerous studies have used it to examine elements such as the challenges and telepresence inherent in gameplay (Bittner and Shipper, 2014; Cai *et al.*, 2022; Pham *et al.*, 2022). These studies have measured dimensions such as engaging challenges and telepresence as prerequisites for achieving Flow. Overall, Flow theory offers a framework for understanding the psychological processes underlying optimal experiences across various contexts, providing valuable insights for enhancing engagement and well-being.

3. Theoretical framework and hypotheses development

Drawing on Flow theory (Csikszentmihalyi, 1975), we developed a conceptual framework to model the intricacies of

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

Flow experiences of consumers derived from video gaming. As depicted in Figure 1, this study conceptualizes the Flow experience in video gaming as a Type II, higher-order reflective-formative latent construct (Guo *et al.*, 2016; Pelet *et al.*, 2017). Our analysis and synthesis of the literature led us to define Flow experience within this context as comprising six key components:

1 concentration;
2 interactivity;
3 telepresence;
4 collaboration;
5 challenge, and
6 goals.

In our theoretical model, Flow experiences are posited to influence the possible outcomes of participating in an activity system of post-purchase gaming, including post-consumption evaluations of customer satisfaction, attitudinal loyalty to the developer, repeat purchase intention and continued interaction intention. These outcomes occur as consequences of Flow experiences in our model.

3.1 Constituent dimensions of flow

3.1.1 Concentration

Pelet *et al.* (2017) posited that Flow is characterized by two fundamental aspects: complete absorption in an activity and the enjoyment derived from it. Similarly, Y. Lu *et al.* (2009) conceptualized Flow as encompassing intrinsic enjoyment, perceived control and focused attention. Achieving a state of Flow requires undivided attention directed toward the activity at hand. Within this framework, concentration in Flow denotes the degree to which an individual's attention is engrossed in the activity, to the exclusion of all else (Pelet *et al.*, 2017). Concentration has emerged as a pivotal element of Flow across several studies (Cai *et al.*, 2022; Yan *et al.*, 2013). In the context of video games, this immersive medium demands complete concentration from players, irrespective of the game's difficulty level (Cowley *et al.*, 2008).

3.1.2 Interactivity

Interactivity has been defined as the dynamic, real-time engagement of participants within a computer-mediated environment (Von der Pütten *et al.*, 2012). Correspondingly, the immediate interactions between individuals and a virtual world are posited to influence the attainment of a Flow state (Animesh *et al.*, 2011). Scholars, such as Nandi *et al.* (2021), have suggested that a platform's capacity to enable users to engage within a virtual environment is indicative of interactivity, potentially leading to positive feedback from customers because of heightened levels of interaction. Further corroborating this notion, Cheng *et al.* (2014) discovered a robust association between interactive experiences and Flow.

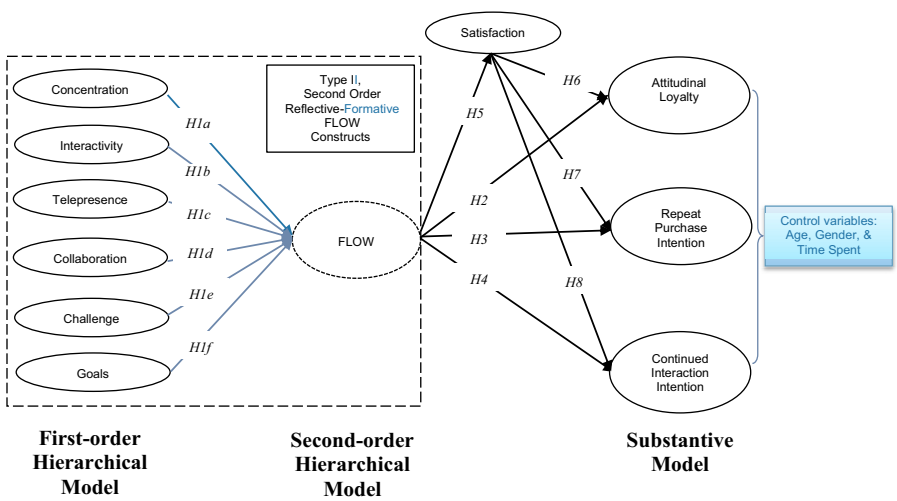
3.1.3 Telepresence

Telepresence, as defined by Hoffman and Novak (1996), refers to the perceptual sensation of being one's virtual self within a virtual environment, facilitated by technology. Within this virtual space, individuals instinctively respond to the virtual environment's elements and surroundings as if they were physically present in the intangible world they are experiencing virtually (Pelet *et al.*, 2017). Studies have revealed a correlation between telepresence and Flow experience in computer-mediated environments (Chen *et al.*, 2000). This correlation is attributed to the displacement of real-world self-awareness that can occur during Flow state, because the virtual environment substitutes for the physical one. Studies on Flow (Nah *et al.*, 2011; Zaman *et al.*, 2010) have also suggested that telepresence can lead to favorable outcomes, such as enhanced enjoyment and concentration.

3.1.4 Collaboration

Collaboration within video gaming denotes the exchange of assistance and support among players within a gaming platform. This collaborative effort involves teaming up with fellow gamers to conquer game objectives or challenges in multiplayer settings (Arakji and Lang, 2007) and sharing gaming techniques or strategies (Voulgari and Komis, 2012), such as exchanging in-game items, resources, or currency and providing guidance,

Figure 1 Conceptual model of the study



Source(s): Authors' own work

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCB

advice or assistance to newer or less experienced players. Such collaborative endeavors foster a sense of belonging to a team or clan (Jimenez *et al.*, 2019) and contribute to mutual support, potentially enhancing interactivity (Von der Pütten *et al.*, 2012). The continuity of interactions can be bolstered when gamers collaborate with others within the platform. Further, collaborative efforts among gamers within a gaming interface can strengthen social bonds (Bluemink *et al.*, 2010), thus solidifying social ties as a crucial foundation for sustained interactions within video games (Molyneux *et al.*, 2015). Therefore, incorporating collaboration as a facet of Flow dimensions is particularly crucial for fortifying the formation of Flow experience within the realm of video gaming.

3.1.5 Challenges

According to Csikszentmihalyi *et al.* (2014), cognitive engagement in challenges or skill development is another crucial determinant of Flow experience. Prior studies have indicated that when game task challenges are appropriately matched with users' skills and surpass a certain threshold of cognitive engagement a state of Flow is triggered (Guo *et al.*, 2016; Pelet *et al.*, 2017). This phenomenon is evident in various contexts, such as human–computer interactions or gamers engagement in online gaming (Dardis *et al.*, 2018; Pham *et al.*, 2022). Video game developers strategically incorporate diverse levels of difficulty within games (Qin *et al.*, 2009) to cater to the varying skill levels of gamers. In certain games, players can select the difficulty level, challenging themselves to tackle more demanding levels. In others, the game presents escalating challenges as players progress. In these two scenarios, individuals' involvement gradually intensifies, leading to heightened enjoyment and prolonged Flow experience (Soutter and Hitchens, 2016).

3.1.6 Goals

As per Guo *et al.* (2016), establishing clear objectives for a task is crucial for attaining optimal performance in any activity. When individuals engage in an activity that has well-defined goals, intense concentration often leads to a state of Flow (Csikszentmihalyi *et al.*, 2014). Studies on online surfing (Chen *et al.*, 2000) and social media engagement (Carlson *et al.*, 2017) underscore the significant impact of clear goals on Flow experience within computer-mediated environments. Similarly, video games are meticulously crafted, and clear objectives are integrated into their core gameplay (Cai *et al.*, 2022). The objectives for players to achieve within video games, as well as players' personal goals, promote entry into a Flow state.

3.2 Conceptualization of flow in video gaming

In understanding Flow, we emphasize its multidimensional nature, especially within the video gaming consumption context. Drawing from methodological frameworks by Lee and Cadogan (2013), Becker *et al.* (2012), and Flow conceptualization by Pelet *et al.* (2017), and Guo *et al.* (2016), we configure Flow as a reflective-formative higher-order construct (Type II). In contrast, Cai *et al.* (2022) in video games and Carlson *et al.* (2017) in social media contexts treat Flow as a reflective-reflective higher-order construct (Type I). However, Siekpe's (2005) study on computer-mediated environments demonstrated that both reflective and formative approaches fit the data similarly when assessing the multidimensionality of Flow.

The formative approach is appropriate here because the six identified components of Flow are distinct and not interchangeable (Becker *et al.*, 2012; Lee and Cadogan, 2013). This conceptualization of Flow, with its distinct components, resonates with Hoffman and Novak's (2009) argument that outcomes of Flow can be influenced by any or all of these components independently. Hence, changes in one component, such as concentration, can occur without necessitating changes in others like perceptions of telepresence or challenges.

These components do not meet the criteria for reflective factor modeling, which assumes they have a high degree of shared variance, or are intercorrelated (Becker *et al.*, 2012; Lee and Cadogan, 2013). Instead, Flow is examined here as a composite of these six specific latent components, which form the Flow construct. Thus, concentration, interactivity, telepresence, collaboration, challenges and goals are formative indicators of video game players' Flow state. As depicted in Figure 1, each of these Flow components is predicted to contribute to and constitute the overarching construct of Flow. Therefore, the Flow state is hypothesized as a latent construct derived from these indicating variables:

H1. Flow comprises six first-order components (*H1a*) Concentration, (*H1b*) Interactivity, (*H1c*) Telepresence, (*H1d*) Collaboration, (*H1e*) Challenge, and (*H1f*) Goals.

3.3 Flow consequences

3.3.1 Attitudinal loyalty

Prior studies on Flow have found that it plays an important role in influencing consumer attitudes (Hoffman and Novak, 2009), including brand attitudes (Aggarwal *et al.*, 2024; Hsu *et al.*, 2012b; Su *et al.*, 2016). This is because individuals experiencing Flow are fully absorbed and engaged in an activity that is inherently enjoyable and this contributes to the development of positive attitudes toward the provider of the experience. The development of positive brand attitudes is strategically important because it is considered the first step in consumers developing preferences for and commitment to a brand, and is a leading indicator of loyalty behavior (Fishbein and Ajzen, 2005; Jacoby and Chestnut, 1978). We argue that more experienced Flow by video game consumers translates to more favorable loyalty attitudes toward video game developers; conversely, less experienced Flow results in less favorable attitudes. Thus, we propose the following hypothesis:

H2. Flow experience during video gaming is positively related to attitudinal loyalty toward video game developers.

3.3.2 Repeat purchase intention

Repeat purchase intention refers to the likelihood of customers continuing to buy from the same producer (Chiu *et al.*, 2012), indicating a positive perception of products and businesses. Additional gaming purchases can include add-ons or downloadable content within already purchased and actively played games. Studies in e-commerce have identified a significant association between Flow state and inclination to

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

repurchase (Lee *et al.*, 2019). The intense pleasure and altered perception of time experienced in a Flow state contributes to the predictability of purchase intentions among customers (Hsu *et al.*, 2012b). Gamers immersed in Flow are anticipated to exhibit stronger impulses and are more likely to make spontaneous purchases related to their gameplay (Lee *et al.*, 2019). Consequently, video game players experiencing Flow are expected to show an increased willingness to make future purchases:

H3. Flow experience during video gaming is positively related to repeat purchase intentions among gamers.

3.3.3 Continued interaction intention

Prior Flow studies have reported that consumers who experience Flow want to replicate or re-experience that state (O'Cass and Carlson, 2010). Therefore, consumers who experience Flow seek to reengage with and revisit the activity that facilitated their Flow experience. In addition, numerous studies have identified a correlation between Flow and exploratory behavior within the Flow activity, across various contexts. For example, in the context of online retail, shoppers experiencing Flow are more inclined to explore product options, engage in comparative evaluations, and seek new information to make informed purchase decisions (Lee *et al.*, 2019). Similarly, in the consumption context of video gaming, players immersed in Flow states are more likely to exhibit exploratory behaviors, such as exploring game levels, experimenting with strategies and seeking new challenges to maintain their engagement and enjoyment (Bittner and Shipper, 2014). Thus, we hypothesize that if video gaming consumers experience Flow, they will continue to use the game. Therefore, we propose the following:

H4. Flow experience during video gaming is positively related with continued interaction intentions among gamers.

3.3.4 Role of satisfaction

Customer satisfaction plays a pivotal role in driving repeat purchase behavior, because it is a prerequisite for an intention to repeatedly use or buy the same products (Hsu *et al.*, 2012b). Mackay *et al.* (2009) suggested that the impulse to repeat purchases stems from customers' positive overall experiences with producers or developers. According to Hsu *et al.* (2012b), satisfaction can mediate the relationship between Flow and purchase impulse, indicating that Flow can lead to satisfaction and contribute to building loyalty toward a game, subsequently influencing customers' underlying purchasing intentions. For video game players, once they develop loyalty to their favorite game, they are more inclined to continue consuming games from the same developer, thereby encouraging subsequent purchases from them. Consequently, we hypothesize the following:

H5. Flow experience during video gaming is positively related with satisfaction among gamers.

H6. Satisfaction with a video game is positively related with attitudinal loyalty toward that game.

H7. Satisfaction with a video game is positively related with repeat purchase intentions for that game.

H8. Satisfaction with a video game is positively related with continued interaction intention with the same video game developer.

4. Methodology

4.1 Sampling

A survey of video gamers was conducted to collect data for this study. The data were obtained from an online questionnaire of adult Chinese university student participants (n = 201) who indicated playing video games for more than a few hours a week. The survey was predominantly distributed through social media and social news sites, such as Weibo, QQ and WeChat, as well as other forums dedicated to games.

Among the respondents, 67.2% were male (M_{age} = 21.73). Approximately 84.6% (170) of respondents were in the 21–25 age group, which is consistent with the median age of video game players in P. R. China (Liu, 2016). Table 1 details the characteristics of our sample. The sample included current undergraduates (44.3%), bachelor's degree holders (52.7%) and graduate-level degree holders (3%). Most participants reported that they played games for up to five hours per week (63.2%); 22.4%, up to 10; and 14.4%, more than 10. Respondents reported playing a variety of game types on different platforms (PC, console, mobile and handheld devices). We classified game types into role-play games (RPG; 10.4%), action games (12.4%), strategy games (32.3%), and other types of games (44.8%). Almost half of respondents (48.3%) used smartphones as their preferred game platform;

Table 1 Profile of respondents (n = 201)

Measure	Items	Frequency	%
Gender	Male	135	67.2
	Female	66	32.8
Age (years)	< 20	24	11.9
	21–25	170	84.6
	> 25	7	3.5
Education	Current undergraduate	89	44.3
	Bachelor degree level	106	52.7
	Graduate degree level	6	3.0
Game type	RPG (role play game)	21	10.4
	Action game	25	12.4
	Strategy game	65	32.3
	Other	90	44.8
Game platform	Mobile game	97	48.3
	Console game	8	4.0
	PC online game	74	36.8
	PC single game	13	6.5
	Handheld game	5	2.5
	Other	4	2.0
Hours spent/week	< 5	127	63.2
	6–10	45	22.4
	> 10	29	14.4

Source(s): Authors' own work

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

personal computer online gaming was the second most preferred (36.8%).

4.2 Measurement

Following protocols by Y.F. Chen and Tjosvold (2005), native speakers who had a good command of English and Chinese translated the survey from the original English version to simplified Chinese. The Flow construct in this study was operationalized as a Type II, reflective-formative construct (Jarvis *et al.*, 2003), whereas the other constructs were reflective constructs. For components of Flow, measures of concentration (four items) and telepresence (three items) were adapted from Pelet *et al.* (2017). Goals and challenges were each measured through three items, adapted to our study context from Guo *et al.* (2016). Interactivity was measured using a scale adapted from Cheng *et al.* (2014) and included three items. Finally, to examine collaboration, four measurement items were adapted from Shi *et al.* (2016). For the outcome variables, three items were used to measure attitudinal loyalty (Cheng *et al.*, 2014) and repeat purchase intention was assessed using two items (Chiu *et al.*, 2012). Continued interaction intention was measured using three items from Shi *et al.* (2016), and satisfaction was measured using three items from Jung (2011). All items were measured on a 7-point scale using 1 = *Strongly disagree* and 7 = *Strongly agree*. Age, gender and time spent playing video games were included as control variables to account for differences among video game players.

Considering the methodological works of Lee and Cadogan (2013) and Flow studies by Pelet *et al.* (2017) and Guo *et al.* (2016), it was appropriate to conceptualize the Flow experience construct as a reflective-formative hierarchical latent variable because, conceptually, the six identified dimensions did not covary, were distinct and were not interchangeable. This idea is analogous to the understanding of Hoffman and Novak (2009), who proposed dimensions of Flow experience, in which an experience may be influenced by one or all Flow dimensions, and a change in one is not necessarily accompanied by changes in the others. For instance, a decrease in a challenge experience can occur separately and would not necessarily be accompanied by a change in interactivity experience, or any of the other salient Flow dimensions perceived by gamers when reflecting on a video game playing experience.

5. Results

5.1 Assessment of measurement model

The model was examined using partial least squares structural equation modeling (PLS-SEM) analysis, using SmartPLS v4.0 (Ringle *et al.*, 2022). PLS-SEM allows for the simultaneous analysis of reflective and second-order formative constructed models using standardized latent variable scores (Hair *et al.*, 2019). This is particularly the case where a second-order formative latent construct (in our case Flow experience) is formed by six first-order reflective latent constructs (i.e. concentration, interactivity, telepresence, collaboration, challenge and goals) that define Flow experience. We modeled the second-order construct by using the hierarchical component model, in which the standardized latent variable scores of the first-order reflective constructs were used as weights for the second-order formative construct (Hair *et al.*, 2024).

A two-step analytical procedure that had direct and indirect effects was performed to test the hypotheses (Hair *et al.*, 2024). A measurement model (see Table 2 for a more detailed analysis) was first assessed, followed by a structural model. The measurement model analysis revealed that every reflective item of all measured constructs had component loadings > 0.75, exceeding the 0.70 threshold, indicating good scale convergence and an average extracted variance for all constructs > 0.64, well above the 0.50 benchmark. All factor loadings had a high value for reliability that met the threshold of 0.70 (Hair *et al.*, 2024).

Discriminant validity was measured by comparing the square root of the average variance extracted (AVE) of the latent variables with the correlations of other constructs (Chin, 1998). The square root AVE scores for attitudinal loyalty, repeat purchase intention and continued interaction intentions were higher than their respective squared inter-correlation values and therefore satisfied the discriminant validity requirements (Fornell–Larcker criterion; Hair *et al.*, 2024). The AVE and correlation matrix in our model constructs demonstrated that all reflective constructs satisfied the Fornell–Larcker criterion for discriminant validity (see Table 3). We also used the heterotrait–monotrait (HTMT) ratio of correlations criterion, where the HTMT criterion should be less than 0.85 (Hair *et al.*, 2024). Our results illustrated that the correlation values corresponding to the respective constructs did not violate HTMT 0.85, indicating that discriminant validity was achieved (see Table 4).

To mitigate the risk of common method bias (CMB), we used randomization in the survey by altering the sequence of measurement items. This approach aimed to prevent respondents from discerning any underlying patterns among the items. To assess CMB, we first conducted a full-collinearity test (Kock, 2015) with a randomly selected dependent variable. The variance inflation factor (VIF) values for all latent constructs ranged from 1.267–2.211, well below the threshold of 3.3. This indicates that CMB is not a significant concern in our analysis. Second, we implemented a single-item marker variable test, as advised by Lindell and Whitney (2001), Murray *et al.* (2011), and Lee *et al.* (2017). This survey item was chosen for its lack of theoretical relevance to the constructs under investigation. Participants rated their agreement with the statement "If someone helps me, I am pleased to help him/her" on a seven-point Likert scale. Correlations were computed between this marker variable and all focal constructs in our study, revealing no significant associations. These findings collectively suggest the absence of CMB in our study.

Flow was identified as a key second-order formative construct, supported by primary first-order reflective constructs. Validating the second-order formative constructs required confirming Outer Weights, Outer Loadings and VIF. Among the proposed dimensions, three outer weights (telepresence ($\beta = 0.54$, $p < 0.01$), concentration ($\beta = 0.29$, $p < 0.01$) and interactivity ($\beta = 0.27$, $p < 0.01$) were statistically significant ($t > 1.96$; $p < 0.01$). Challenge was not- a significant indicator ($\beta = 0.19$, $p > 0.05$), while two outer weights collaboration ($\beta = 0.05$, $p > 0.05$) and goals ($\beta = 0.12$, $p > 0.05$)

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

Table 2 Measurement model

Components and manifest variables	Loading (*t*-value)
Concentration: (AVE: 0.81, CR (rho_a): 0.92, α: 0.92)	
I am deeply engrossed in playing the video game	0.87 (46.89)*
I am absorbed intensely in playing the video game	0.89 (45.50)*
My attention is focused on playing the video game	0.91 (43.28)*
I concentrate fully on playing the video game	0.90 (48.91)*
Interactivity: (AVE: 0.74, CR (rho_a): 0.83, α: 0.82)	
When I play video games there is very little waiting time between my actions and the device's response	0.88 (30.80)*
Interacting with the video game is fast	0.91 (47.95)*
The video games I play usually load quickly	0.82 (24.67)*
Telepresence: (AVE: 0.77, CR (rho_a): 0.86, α: 0.85)	
I forget my immediate environment when I play video games	0.89 (47.23)*
While playing video games, I feel I am in the world the video game created	0.90 (56.07)*
While playing video games, the game comes to me and creates a new world for me, and this world suddenly disappears when I stop playing.	0.84 (22.79)*
Collaboration: (AVE: 0.74, CR (rho_a): 0.89, α: 0.88)	
In this video game, I can help other players	0.84 (28.15)*
In this video game, I can receive help from other players	0.84 (21.85)*
In this video game, I can provide information to other players	0.89 (40.00)*
In this video game, I can share my ideas with other players	0.86 (28.62)*
Challenge: (AVE: 0.71, CR (rho_a): 0.79, α: 0.79)	
I am challenged by the video game tasks, but I believe my skills allow me to meet the challenge in the video game	0.81 (18.77)*
My abilities match the high challenge of the task in the video game	0.87 (30.86)*
I feel I am competent enough to meet the high demands of the task in the video game	0.84 (28.65)*
Goals: (AVE: 0.72, CR (rho_a): 0.81, α: 0.81)	
I know clearly what I want to do in the video game	0.87 (35.64)*
I have a strong sense of what I want to do in the video game	0.87 (31.00)*
I know what I want to achieve in the video game	0.81 (25.47)*
Satisfaction: (AVE: 0.80, CR (rho_a): 0.88, α: 0.87)	
Playing this video game makes me feel very satisfied	0.91 (49.28)*
Playing this video game makes me feel very pleased	0.87 (29.76)*
I am content with this video game	0.90 (51.02)*
Attitudinal Loyalty (AVE: 0.64, CR (rho_a): 0.73, α: 0.72)	
I would recommend this game or games from the developer to others	0.82 (22.79)*
I consider this video game to be my first choice to play	0.82 (21.48)*
I would like to play this video game in the future	0.75 (14.31)*
Repeat Purchase Intention (RPI): (AVE: 0.86, CR (rho_a): 0.84, α: 0.84)	
I consider the developer of this video game to be my first choice for buying new games in the future	0.93 (60.42)*
I will likely continue buying products from this video game developer in the future	0.93 (61.26)*
Continued Use Intention: (AVE: 0.71, CR (rho_a): 0.71, α: 0.73)	
I want to spend more time playing this video game	0.92 (42.27)*
I will try similar video games from this game developer in the future	0.76 (12.57)*

Note(s): AVE = average variance extracted; CR = composite reliability; α = Cronbach's alpha; * = significant at $p < 0.01$
Source(s): Authors' own work

were also found to be non-significant, as detailed in Table 5 and Figure 2.

It's important to note that non-significant indicator weights should not necessarily imply poor measurement quality (Hair *et al.*, 2024). Researchers should also assess the absolute contribution of each formative indicator independently, indicated by their outer loadings – the unique information they contribute to their constructs (Hair *et al.*, 2024; Sarstedt *et al.*, 2019). In this study, all outer loadings exceeded 0.50 for each first-order construct (Sarstedt *et al.*, 2019), thus we retain collaboration, challenges and goals in the second-order construct of Flow in our measurement model. The evaluation of collinearity of the Flow indicators found that the VIF values were all below the recommended

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

Table 3 Discriminant validity using Fornell–Larcker criterion

	1	2	3	4	5	6	7	8	9	10
1. Attitudinal loyalty	*0.79*	–	–	–	–	–	–	–	–	–
2. Challenge	0.29	*0.84*	–	–	–	–	–	–	–	–
3. Collaboration	0.29	0.28	*0.85*	–	–	–	–	–	–	–
4. Concentration	0.43	0.43	0.38	*0.89*	–	–	–	–	–	–
5. Continued use intention	0.52	0.31	0.17	0.47	*0.84*	–	–	–	–	–
6. Goals	0.39	0.65	0.41	0.52	0.28	*0.85*	–	–	–	–
7. Interactivity	0.32	0.39	0.23	0.33	0.19	0.44	*0.86*	–	–	–
8. Repeat purchase intention	0.45	0.25	0.18	0.37	0.56	0.26	0.22	*0.93*	–	–
9. Satisfaction	0.66	0.41	0.27	0.38	0.46	0.46	0.44	0.45	*0.89*	–
10. Telepresence	0.42	0.17	0.20	0.45	0.50	0.25	0.11	0.40	0.32	*0.87*

Note(s): Diagonal italic entries are the square root of AVE; others are correlations
Source(s): Authors' own work

Table 4 Discriminant validity using heterotrait–monotrait ratio (HTMT)

	1	2	3	4	5	6	7	8	9	10
1. Attitudinal loyalty	–	–	–	–	–	–	–	–	–	–
2. Challenge	0.39	–	–	–	–	–	–	–	–	–
3. Collaboration	0.37	0.33	–	–	–	–	–	–	–	–
4. Concentration	0.53	0.50	0.42	–	–	–	–	–	–	–
5. Continued use intention	0.75	0.43	0.20	0.59	–	–	–	–	–	–
6. Goals	0.51	0.81	0.49	0.60	0.39	–	–	–	–	–
7. Interactivity	0.41	0.49	0.26	0.38	0.24	0.54	–	–	–	–
8. Repeat purchase intention	0.56	0.31	0.20	0.42	0.78	0.32	0.27	–	–	–
9. Satisfaction	0.82	0.50	0.30	0.42	0.60	0.55	0.52	0.53	–	–
10. Telepresence	0.53	0.20	0.23	0.51	0.65	0.30	0.12	0.47	0.36	–

Source(s): Authors' own work

Table 5 Second-Order Construct (flow) validity

SOC	FOCs	Outer weights	*t*-statistics	*p*-values	Outer loadings	VIF
Flow	*H1a*: Concentration	0.287	2.548	0.011	0.780	1.732
	H1b: Interactivity	0.265	3.154	0.002	0.560	1.301
	H1c: Telepresence	0.539	5.151	0.000	0.771	1.266
	H1d: Collaboration	0.048	0.521	0.602	0.532	1.270
	H1e: Challenge	0.190	1.653	0.098	0.602	1.824
	H1f: Goals	0.115	0.919	0.358	0.666	2.201

Note(s): SOC = second-order construct; FOCs = first-order constructs; VIF = variance inflation factor
Source(s): Authors' own work

threshold of 5 (Hair *et al.*, 2024). Meeting these criteria established the second-order validity of the formative Flow construct in our study. Taken collectively, the results provide partial support for the first hypothesis (*H1a-f*). In this study Flow is a second-order construct that is formatively shaped by three first-order constructs: concentration, interactivity, telepresence.

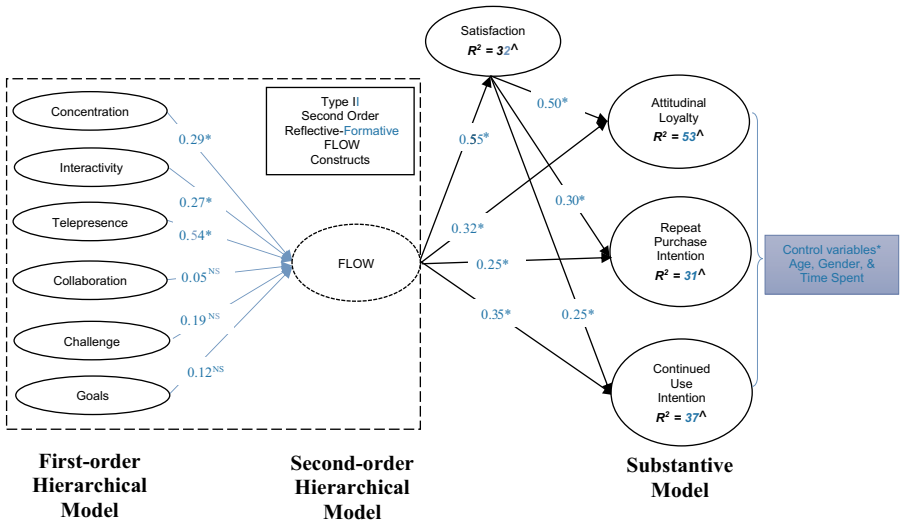
Before testing the structural model, we checked the measurement model fit using standardized root mean square residual (SRMR; Hu and Bentler, 1999). Our result demonstrated that the model fit was good; with an SRMR composite factor model of 0.09, which sits below the 0.10 benchmark (Hair *et al.*, 2024; Hu and Bentler, 1999). Controlling for age, gender and time spent playing video games did not result in any significant changes to the model, regardless of their inclusion or exclusion. Please see Table 6 for detailed results.

5.2 Assessment of structural model

To test the structural model, we examined the size and significance of the path coefficients. As presented in Figure 2 and Table 6, all paths were significant ($p < 0.01$), supporting all of our hypotheses.

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

Figure 2 Structural model



Note(s): * = Significant at $p < 0.01$; NS = Non-significant
Source(s): Authors' own work

Table 6 Hypotheses results for the structural model

Hypothesis	Predicted constructs	Predictor constructs	R^2	Path (ß)	*t*-statistics	Conclusion
H2	Attitudinal loyalty	Flow	0.53^{+}	0.32	4.45*	Supported
H3	Repeat purchase intention	"	0.31^{+}	0.25	2.96*	Supported
H4	Continued use intention	"	0.37^{+}	0.35	4.28*	Supported
H5	Satisfaction	"	0.32^{+}	0.55	8.45*	Supported
H6	Attitudinal loyalty	Satisfaction	–	0.50	7.86*	Supported
H7	Repeat purchase intention	"	–	0.30	4.47*	Supported
H8	Continued use intention	"	–	0.25	3.23*	Supported
Controls variables						
	Attitudinal loyalty	Age	–	0.09	1.11^{ns}	
	Repeat purchase intention	"	–	0.08	1.56^{ns}	
	Continued use intention	"	–	0.05	0.70^{ns}	
	Attitudinal loyalty	Gender	–	0.03	0.42^{ns}	
	Repeat purchase intention	"	–	−0.27	4.53*	
	Continued use intention	"	–	−0.23	3.10*	
	Attitudinal loyalty	Time spent	–	−0.01	0.12^{ns}	
	Repeat purchase intention	"	–	0.07	1.08^{ns}	
	Continued use	"	–	0.01	1.06^{ns}	

Note(s): * = significant at $p < 0.01$; ^{ns} = nonsignificant; $^{+}$ = exceeds criterion of 0.19; average variance accounted (AVA): 0.38
Source(s): Authors' own work

The hypothesized relationship between Flow and satisfaction was supported (*H5*; $\beta = 0.55$; $p < 0.01$). This positive relationship was stronger than the relationship between Flow and attitudinal loyalty (*H2*; $\beta = 0.32$; $p < 0.01$), repeat purchase intention (*H3*; $\beta = 0.25$; $p < 0.01$) and continued use intention (*H4*; $\beta = 0.35$; $p < 0.01$). Moreover, satisfaction exhibited a stronger relationship with attitudinal loyalty (*H6*; $\beta = 0.50$; $p < 0.01$) than with repeat purchase intention (*H7*; $\beta = 0.30$; $p < 0.01$) and continued use intention (*H8*; $\beta = 0.25$; $p < 0.01$). In addition, Flow experience accounted for 32% of the variance in satisfaction ($R^2 = 0.32$); 53% of attitudinal loyalty ($R^2 = 0.53$); 37% of continued use intention ($R^2 = 0.37$); and 31% of repeat purchase intention ($R^2 = 0.31$). Key benchmarks were taken from Chin (1998), who stated that R^2 values must be greater than 0.19, providing evidence of the significance of these relationships in our results.

We next examined the predictive relevance of our model by checking Stone-Geisser's Q-square value (Hair *et al.*, 2024). This also provides an indication of how well the observed values are reconstructed by the model (Chin, 1998). Following the general

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCB

principle, a Q^2 greater than zero signifies that a model has predictive relevance (Hair *et al.*, 2024). In our substantive model, all four dependent variables indicated acceptable predictive relevance. Subsequently, the effect size (f^2) assesses the influence of an exogenous latent variable on an endogenous one through R^2 changes. Values of 0.02, 0.15 and 0.35 indicate small, medium and large effects, respectively (Hair *et al.*, 2024). In our model, effect sizes for satisfaction, attitudinal loyalty, repeat purchase intentions and continued interaction intentions ranged from 0.054–0.429, meeting established benchmarks. Thus, all of the results supported the conceptualized model.

5.3 Post hoc mediation analysis

Although we did not formulate this in a hypothesis, we investigated how overall satisfaction mediates the relationship between Flow experience and attitudinal loyalty, repeat purchase intentions and continued interaction intentions. Subsequently, we conducted mediation analysis to clarify the nonsignificant direct roles, following Hair *et al.*'s (2024) framework, which encompasses complementary partial mediation, competitive partial mediation and full mediation using SmartPLS. The confirmation of mediation requires a significant specific indirect effect. Direct-only non-mediation occurs when the direct effect is significant, and no-effect non-mediation occurs when there is no direct effect. Full mediation is observed when the specific indirect effect is significant, and the direct effect is not, and partial mediation is evidenced in the model by the simultaneous presence of a significant direct effect and a significant indirect effect (Hair *et al.*, 2024). Thus, our analysis, as illustrated in Table 7, shows that satisfaction is a partial mediator in the relationship between Flow experience and attitudinal loyalty, repeat purchase intentions and repeat purchase intentions.

6. Discussion and implications

This study enriches the consumer marketing literature by investigating the factors that are likely to influence Chinese video gamers' loyalty formation via game-based Flow experiences. This study holds noteworthy scholarly and practical implications offering valuable insights into the mechanisms for inducing Flow in gaming experiences to explain consumer behavior which are discussed next.

6.1 Implications for theory

From a theoretical perspective, this study makes several important contributions by offering a nuanced understanding of the Flow experience in the context of video gaming. While Flow remains a universal construct, its manifestations may vary substantially depending on variables related to the task eliciting the Flow state. In the video game setting, this includes the interplay between game design and player-related variables, such as expectations.

The findings of this study revealed that the Flow dimensions of goals, challenges and collaboration were not statistically significant in fostering Flow experiences for Chinese gamers. This outcome can be attributed to the distinctive characteristics of the Chinese gaming market, which is dominated by mobile and casual games. These games are designed to prioritize ease of access, quick engagement and instant gratification, often with simplified mechanics that do not rely heavily on long-term goal setting, complex challenges or extensive collaboration. Consequently, these dimensions may hold less relevance for the primary audience and game types within this market.

However, the relative insignificance of these dimensions does not diminish their importance across other contexts. For example, goal orientation and challenge may play a pivotal role in genres such as role-playing games (RPGs) or massively multiplayer online games, where player engagement often hinges on achieving milestones and overcoming progressive challenges. Similarly, collaboration becomes a core driver of Flow in team-based games, such as multiplayer online battle arenas and esports titles, where cooperative gameplay and team dynamics shape the overall experience. These findings suggest that the pathways to achieving Flow are context dependent, influenced by factors such as game type, audience preferences and cultural norms. This aligns with broader research emphasizing the adaptability of Flow theory across diverse domains (Stremersch *et al.*, 2023).

The study also provides empirical support to existing consumer marketing literature on the nature of Flow concerning its multidimensional properties as a reflective-formative construct (e.g. Guo *et al.*, 2016; Pelet *et al.*, 2017). These findings clarify how underlying psychological processes collectively contribute to the construction of a Flow state for consumers during video gaming, offering new insights into its applicability across contexts.

Accordingly, this study addresses a critical gap in the literature by applying Flow theory to the multi-device video game market in China, a unique environment and market that remains underexplored. By focusing on the six dimensions of Flow – concentration, interactivity, telepresence, collaboration, challenge and goals – this research enhances our understanding of how these elements impact player engagement. It identifies telepresence, concentration and interactivity as pivotal dimensions.

Further, this study advances Flow theory by demonstrating its impact on satisfaction, repeat purchase intentions, continued interaction intentions and attitudinal loyalty toward a game developer. Notably, the study provides the first

Table 7 Satisfaction as mediator analysis results

Paths	Specific indirect effects	Direct effects	Total effects	Types of mediation
Flow → satisfaction → repeat purchase intention	0.165*	0.250*	0.415*	Partial mediation
Flow → satisfaction → attitudinal loyalty	0.275*	0.320*	0.595*	Partial mediation
Flow → satisfaction → continued use intention	0.138*	0.350*	0.488*	Partial mediation

Note(s): *$p < 0.01$
Source(s): Authors' own work

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

empirical evidence demonstrating satisfaction does partially mediate Flow and these outcomes. In each of these relationships, the effect of Flow is partially mediated by consumer satisfaction. This highlights the dual importance of intrinsic enjoyment (Flow) and extrinsic motivators (satisfaction), providing a nuanced understanding of how Flow influences behavioral intentions.

These findings confirm prior research suggesting that satisfaction partially mediates the positive effects of Flow on other consumer attitudes and behaviors (Cai *et al.*, 2022; Chen, Lu, and Luor, 2018b; Hsu *et al.*, 2012a; Mackay *et al.*, 2009). They extend theoretical models by elucidating the complex interplay between Flow, satisfaction and player outcomes, challenging existing frameworks and calling for further exploration of alternative mechanisms that drive consumer behavior in gaming contexts.

Overall, this research contributes to the theoretical framework of Flow theory by providing empirical evidence of its multi-dimensional nature and contextual applicability. It advances the discourse by refining our understanding of how Flow dimensions function in diverse gaming markets and offering actionable insights into their influence on consumer behavior. This study lays the groundwork for future research to further explore the nuanced relationships between Flow, satisfaction and behavioral intentions within video gaming.

6.2 Implications for practice

This study offers actionable insights for game developers and marketers seeking to enhance consumer engagement through the cultivation of Flow experiences. More specific to this context, the importance of prioritizing Flow dimensions that were shown to significantly contribute to Flow – *concentration*, interactivity and *telepresence*. While the study did not find statistical significance for goals, challenges and collaboration as contributors to Flow in this context, this does not diminish their potential relevance across all gaming genres or audiences. Instead, it suggests that the influence of these Flow dimensions may vary depending on the type of game and the target demographic.

To enable Flow, game designers should focus on high-quality visuals, engaging soundscapes and compelling narratives that foster concentration and a deep sense of telepresence. Clear feedback systems, adaptive challenges and intuitive interactivity further enhance Flow, creating a seamless and enjoyable gameplay experience. Distractions, such as cumbersome controls or excessive downtime, should be minimized to sustain players' focus.

Although goals and collaboration were not shown as significant contributors in this study, developers should recognize their potential importance in specific gaming contexts. For example, goals may play a central role in games where progression and achievement are core to the player experience, such as strategy or RPG. Similarly, collaboration may hold greater significance in multiplayer and team-based games, where social interaction and teamwork are essential to gameplay. Designers should carefully evaluate the specific needs of their target audience and adapt Flow-inducing features accordingly.

For marketers, the findings highlight the value of promoting the most impactful Flow dimensions that are relevant to their market. This study guides the Chinese and mobile gaming market specifically, and those of similar flow dimensional makeup. Accordingly, marketing campaigns should emphasize the immersive aspects of games, such as their ability to captivate attention (*concentration*), challenge players at optimal difficulty levels (*challenges*) and foster a strong sense of presence within the game world (*telepresence*). Features that enhance interactivity, such as responsive controls or adaptive environments, should also be prominently showcased.

When considering the non-significant findings for *goals* and *collaboration*, marketers should adapt their messaging based on the game's specific features and intended audience. For example, in games where collaboration is a secondary feature, promotional efforts may focus more on individual player achievements and immersive storytelling. Conversely, in multiplayer or co-op games where collaboration is critical, campaigns can emphasize teamwork and community-building aspects, even if such dimensions were not statistically supported in the current study.

The nuanced results of this study suggest that Flow dimensions may not be universally applicable across all gaming contexts. For casual or mobile games, where ease of play is prioritized, goals and challenges might be de-emphasized in favor of intuitive mechanics and quick gratification. Meanwhile, hardcore gaming audiences, such as those who engage with competitive esports or MMORPGs, may place greater importance on collaboration and goal setting.

Finally, the implications of these findings extend to cultural and regional differences in gaming preferences. The dimensions of *goals* and *collaboration* may be more relevant in some cultural contexts than others, and game developers should remain mindful of these variations. For instance, Western audiences may emphasize individual achievement, while players in Asian markets may value team-oriented gameplay and cooperative elements.

In conclusion, while not all Flow dimensions were statistically supported in this study, their relevance should be evaluated within the context of specific games, genres and target markets. By leveraging these insights, game developers and marketers can design and promote games that resonate with their intended audiences, driving deeper engagement and long-term player loyalty.

6.3 Limitations and future directions

In this section, we address the inherent constraints of our study and propose directions for future researchers to advance understanding in the field of video game consumer behavior. Building on insights derived from the study, we present Table 8, outlining potential research topics, associated research questions and managerial challenges. This structured approach identifies key areas for further investigation, offering a roadmap for scholars and practitioners to navigate the complex landscape of video game consumer behavior.

A common constraint in this study, typical of cross-sectional online surveys, is the reliance on self-reported data. Supplementing survey data with video game log data would enhance the robustness of our measures and model. Access to game log data would unveil the interplay between players' skills and knowledge, providing a more precise assessment of Flow experience. Second, this study encompassed various platforms

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCB

Table 8 Future research directions advancing flow experience in video gaming

Research topic	Research questions	Managerial challenge
Integration of game log data	How does the interplay between players' skills and knowledge affect the flow experience in video gaming?	Supplementing survey data with actual video game log data would enhance the robustness of measures and models. Access to game log data would unveil the interplay between players' skills and knowledge, providing a more precise assessment of the flow experience
Experimental designs in flow research	How do specific game elements, such as difficulty levels or narrative complexity, influence flow experiences? How do these experiences influence subsequent behaviour within the game? What is the impact of social interaction within gaming environments on players' flow experiences, and how does this influence their purchasing decisions and overall satisfaction with the gaming experience?	Manipulating specific game elements in experiments can provide insights into their impact on players' immersion and engagement with the game. Social gaming experiments offer opportunities to explore the role of social interaction in facilitating flow and influencing purchasing decisions
Investigating flow as a risk factor	What role do intense and prolonged flow states play in addictive gaming behaviours and excessive spending?	Targeted interventions and prevention strategies can be developed to promote responsible gaming practices and mitigate gaming addiction's adverse effects
Categorisation of video game genres	How do different game genres influence flow experiences and consumer behaviour? To what extent do these experiences vary across gaming platforms?	The study may not encompass all types of games, and certain components may not apply to specific gaming platforms. Future studies should categorise game genres and focus primarily on specific categories of games to provide a comprehensive exploration of the subject matter
Impact of virtual reality on flow experiences	How do immersive spatial technology-enabled gaming environments (virtual reality, extended technologies) affect flow experiences? What is the influence on consumer behaviour? What are the negative effects of flow experiences in virtual reality-based gaming environments (e.g. motion sickness, escapism and avoidance, addiction and compulsive gaming)?	The emergence of virtual reality and extended reality technology offers new opportunities for enhancing gaming experiences. Understanding the unique aspects of spatial computing-based gaming can inform the development of immersive and engaging gaming experiences
Role of artificial intelligence in game design	How does the integration of artificial intelligence technologies affect flow experiences in video gaming?	Artificial intelligence technologies are increasingly being used in game design to enhance player experiences. Examining artificial intelligence-driven game elements, such as adaptive difficulty levels or personalised content recommendations, can provide insights into how these technologies influence player engagement
Cross-cultural comparison of flow experiences	How do cultural differences influence the perception and experience of flow in gaming, and how do these differences affect consumer behaviour across diverse cultural contexts? What are the key cultural factors that shape gaming preferences, motivations and behaviours, and how can game developers and marketers effectively tailor their strategies to target diverse audiences in different cultural settings?	Cultural factors may affect players' perceptions and experiences of flow in gaming. Comparing gaming preferences, motivations and behaviours across cultural contexts can provide valuable insights for game developers and marketers targeting diverse audiences

Source(s): Authors' own work

and gaming contexts, yet future researchers should investigate specific game types across platforms, exploring how varying degrees of influence affect Flow experiences. Such investigations could focus on the role of artificial intelligence and emerging spatial technology-enabled gaming environments (e.g. virtual/extended reality headsets by Meta Quest 3 and Apple Vision Pro) and their ability to sustain Flow experiences that affect consumer behavior in gaming.

Third, this study adopted a survey approach in data collection to understand Flow experiences and their impact on consumer behavior in video gaming. Future researchers should consider experimental designs, which have received far less attention in literature. One promising avenue involves manipulating specific game elements to observe their effects on Flow states and subsequent consumer actions. For instance, researchers could vary elements such as difficulty levels, narrative complexity or the

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

presence of social features to discern their impact on players' immersion and engagement. Moreover, social gaming experiments provide a valuable opportunity to explore how social interaction within gaming environments affects Flow experiences and consumer behavior. By manipulating factors such as team dynamics, communication methods and collaboration strategies, researchers can gain insights into the role of social aspects in facilitating Flow and influencing players' purchasing decisions.

Integrating biometric and neurological measurements into experimental designs also allows for a deeper understanding of the physiological and neurological correlates of Flow experiences during gaming. By tracking metrics such as electroencephalography signals or heart rate variability, researchers can uncover the underlying mechanisms of Flow and how they relate to consumer behavior, such as purchase intentions or brand preferences. These experimental approaches can provide a nuanced understanding of the interplay between Flow experiences in video games and subsequent consumer actions, informing marketing strategies and game design practices.

Fourth, future studies on Flow should examine its impact on behaviors detrimental to gamers by investigating the potential role of Flow experiences as protective and risk factors. This issue is particularly relevant for marketing practitioners in video gaming to address Sustainable Development Goal 12, which focuses on ensuring sustainable consumption and production patterns. This includes the development of responsible marketing practices aimed at promoting sustainable consumption practices and fostering individual wellbeing. Studies could explore how intense and prolonged Flow states contribute to addictive gaming behaviors, including excessive play, compulsive spending on in-game purchases and engagement in harmful activities, such as cyberbullying or aggressive behavior toward other players. Further, researchers should examine the underlying psychological mechanisms that mediate the relationship between Flow experiences and negative gaming outcomes, considering factors such as impulsivity, self-control and psychological wellbeing. By understanding the complex interplay between Flow experiences and problematic gaming behaviors, researchers can develop targeted interventions and prevention strategies to promote responsible gaming practices and mitigate the adverse effects of gaming addiction and excessive in-game purchasing on gamers' mental health and wellbeing.

Finally, this study examined a broad spectrum of video games, and theories adapted from Flow studies were tailored to fit the video game landscape in China. However, given the vast diversity within the realm of video games, certain aspects of our inquiries may not encompass all types of games and certain components may not apply to specific gaming platforms. Consequently, our survey may not provide a comprehensive exploration of the subject matter, and future studies should categorize different game genres and primarily concentrate on a specific category of games. Regardless of these limitations, this study has contributed to the video game and Flow experience body of knowledge. Future research in the video game market is needed, as much is yet unknown about video game consumption, Flow and the role of satisfaction in video game consumption outcomes.

References

Aggarwal, E., Singh, A.B. and Misra, R. (2024), "Does consumption values and ascribed responsibility predict attitudes towards sustainable luxury brands", *Journal of Consumer Marketing*, Vol. 41 No. 2, pp. 180-195, doi: 10.1108/JCM-05-2023-6058.

Animesh, A., Pinsonneault, A., Yang, S.B. and Oh, W. (2011), "An odyssey into virtual worlds: exploring the impacts of technological and spatial environments on intention to purchase virtual products", *MIS Quarterly*, Vol. 35 No. 3, pp. 789-810, doi: 10.2307/23042809.

Arakji, R.Y. and Lang, K.R. (2007), "Digital consumer networks and producer-consumer collaboration: innovation and product development in the video game industry", *Journal of Management Information Systems*, Vol. 24 No. 2, pp. 195-219, doi: 10.2753/MIS0742-1222240208.

Becker, J.M., Klein, K. and Wetzels, M. (2012), "Hierarchical latent variable models in PLS-SEM: guidelines for using reflective-formative type models", *Long Range Planning*, Vol. 45 Nos 5/6, pp. 359-394.

Bittner, J.V. and Shipper, J. (2014), "Motivational effects and age differences of gamification in product advertising", *Journal of Consumer Marketing*, Vol. 31 No. 5, pp. 391-400, doi: 10.1108/JCM-04-2014-0945.

Bluemink, J., Hämäläinen, R., Manninen, T. and Järvelä, S. (2010), "Group-level analysis on multiplayer game collaboration: how do the individuals shape the group interaction?", *Interactive Learning Environments*, Vol. 18 No. 4, pp. 365-383, doi: 10.1080/10494820802602444.

Cai, X., Cebollada, J. and Cortiñas, M. (2022), "Self-report measure of dispositional flow experience in the video game context: conceptualisation and scale development", *International Journal of Human-Computer Studies*, Vol. 159, doi: 10.1016/j.ijhcs.2021.102746.

Carlson, J., de Vries, N.J., Rahman, M.M. and Taylor, A. (2017), "Go with the flow: engineering flow experiences for customer engagement value creation in branded social media environments", *Journal of Brand Management*, Vol. 24 No. 4, pp. 334-348, doi: 10.1057/s41262-017-0054-4.

Chen, Y.F. and Tjosvold, D. (2005), "Cross-cultural leadership: goal interdependence and leader–member relations in foreign ventures in China", *Journal of International Management*, Vol. 11 No. 3, pp. 417-439, doi: 10.1016/j.intman.2005.06.007.

Chen, Y.M., Hsu, T.H. and Lu, Y.J. (2018a), "Impact of flow on mobile shopping intention", *Journal of Retailing and Consumer Services*, Vol. 41, pp. 281-287, doi: 10.1016/j.jretconser.2017.04.004.

Chen, C.S., Lu, H.P. and Luor, T. (2018b), "A new flow of location based service mobile games: non-stickiness on pokémon go", *Computers in Human Behavior*, Vol. 89, pp. 182-190, doi: 10.1016/j.chb.2018.07.023.

Chen, H., Wigand, R.T. and Nilan, M. (2000), "Exploring web users' optimal flow experiences", *Information Technology & People*, Vol. 13 No. 4, pp. 263-281, doi: 10.1108/09593840010359473.

Cheng, L-K., Chieng, M-h. and Chieng, W-h (2014), "Measuring virtual experience in a three-dimensional virtual reality interactive simulator environment: a structural

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCl

equation modeling approach", *Virtual Reality*, Vol. 18 No. 3, pp. 173-188, doi: 10.1007/s10055-014-0244-2.

Chin, W.W. (1998), "The partial least squares approach to structural equation modeling", *Modern Methods for Business Research*, Vol. 295 No. 2, pp. 295-336.

Chiu, C.M., Hsu, M.H., Lai, H. and Chang, C.M. (2012), "Re-examining the influence of trust on online repeat purchase intention: the moderating role of habit and its antecedents", *Decision Support Systems*, Vol. 53 No. 4, pp. 835-845, doi: 10.1016/j.dss.2012.05.021.

Cowley, B., Charles, D., Black, M. and Hickey, R. (2008), "Toward an understanding of flow in video games", *Computers in Entertainment*, Vol. 6 No. 2, pp. 1-27, doi: 10.1145/1371216.1371223.

Csikszentmihalyi, M. (1975), *Beyond Boredom and Anxiety*, Jossey-Bass Publishers, San Francisco, CA.

Csikszentmihalyi, M. (1990), *Flow: The Psychology of Optimal Experience*, Harper & Row, New York, NY.

Csikszentmihalyi, M., Abuhamdeh, S. and Nakamura, J. (2014), '*Flow*', *Flow and the Foundations of Positive Psychology: The Collected Work of Mihaly Csikszentmihalyi*, Springer, Dordrecht, pp. 227–238, doi: 10.1007/978-94-017-9088-8_15.

Dardis, F., Schmierbach, M., Sherrick, B. and Luckman, B. (2018), "How game difficulty and ad framing influence memory of in-game advertisements", *Journal of Consumer Marketing*, Vol. 36 No. 1, pp. 1-11, doi: 10.1108/JCM-07-2016-1878.

Fishbein, M. and Ajzen, I. (2005), "Theory-based behavior change interventions: comments on hobbis and sutton", *Journal of Health Psychology*, Vol. 10 No. 1, pp. 27-31, doi: 10.1177/1359105305048552.

GameCompanies (2023), "Chinese game industry", available at: https://gamecompanies.com/industries/asia/chinese-game-industry/companies (accessed 25 December 2023).

Guo, Z., Xiao, L., Van Toorn, C., Lai, Y. and Seo, C. (2016), "Promoting online learners' continuance intention: an integrated flow framework", *Information & Management*, Vol. 53 No. 2, pp. 279-295, doi: 10.1016/j.im.2015.10.010.

Hair, J.F., Risher, J.J., Sarstedt, M. and Ringle, C.M. (2019), "When to use and how to report the results of PLS-SEM", *European Business Review*, Vol. 31 No. 1, pp. 2-24, doi: 10.1108/EBR-11-2018-0203.

Hair, J.F., Sarstedt, M., Ringle, C.M. and Gudergan, S.P. (2024), *Advanced Issues in Partial Least Squares Structural Equation Modeling*, (2nd edition) Sage, Thousand Oaks, CA.

Hew, J.J., Lee, V.H., T'ng, S.T., Tan, G.W.H., Ooi, K.B. and Dwivedi, Y.K. (2023), "Are online mobile gamers really happy? On the suppressor role of online game addiction", *Information Systems Frontiers*, Vol. 26 No. 1, pp. 217-249, doi: 10.1007/s10796-023-10377-7.

Hoffman, D.L. and Novak, T.P. (1996), "Marketing in hypermedia computer-mediated environments: conceptual foundations", *Journal of Marketing*, Vol. 60 No. 3, pp. 50-68, doi: 10.1177/002224299606000304.

Hoffman, D.L. and Novak, T.P. (2009), "Flow online: lessons learned and future prospects", *Journal of Interactive Marketing*, Vol. 23 No. 1, pp. 23-34, doi: 10.1016/j.intmar.2008.10.003.

Hsu, C.L., Chang, K.C. and Chen, M.C. (2012a), "Flow experience and internet shopping behavior: investigating the moderating effect of consumer characteristics", *Systems Research and Behavioral Science*, Vol. 29 No. 3, pp. 317-332, doi: 10.1002/sres.1101.

Hsu, C.L., Chang, K.C. and Chen, M.C. (2012b), "The impact of website quality on customer satisfaction and purchase intention: perceived playfulness and perceived flow as mediators", *Information Systems and e-Business Management*, Vol. 10 No. 4, pp. 549-570, doi: 10.1007/s10257-011-0181-5.

Hu, L.T. and Bentler, P.M. (1999), "Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives", *Structural Equation Modeling: A Multidisciplinary Journal*, Vol. 6 No. 1, pp. 1-55, doi: 10.1080/10705519909540118.

Huang, T.L., Wu, C.N., Chang, M.H., Liao, G.Y. and Teng, C.I. (2022), "From skill growth expectancy to online game commitment", *Computers in Human Behavior*, Vol. 137, p. 107422, doi: 10.1016/j.chb.2022.107422.

Jacoby, J. and Chestnut, R.W. (1978), *Brand Loyalty Measurement and Management*, John Wiley & Sons, New York, NY.

Jarvis, C., Mackensie, S. and Podsakoff, P. (2003), "A critical review of construct indicators and measurement model misspecification in marketing and consumer research", *Journal of Consumer Research*, Vol. 30 No. 2, pp. 199-218, doi: 10.1086/376806.

Jimenez, N., San-Martin, S., Camarero, C. and Cabezudo, R. S.J. (2019), "What kind of video gamer are you?", *Journal of Consumer Marketing*, Vol. 36 No. 1, pp. 218-227, doi: 10.1108/JCM-06-2017-2249.

Jung, Y. (2011), "Understanding the role of sense of presence and perceived autonomy in users' continued use of social virtual worlds", *Journal of Computer-Mediated Communication*, Vol. 16 No. 4, pp. 492-510, doi: 10.1111/j.1083-6101.2011.01540.x.

Kock, N. (2015), "Common method bias in PLS-SEM: a full collinearity assessment approach", *International Journal of e-Collaboration*, Vol. 11 No. 4, pp. 1-10.

Kuo, A., Lutz, R.J. and Hiler, J.L. (2016), "Brave new world of Warcraft: a conceptual framework for active escapism", *Journal of Consumer Marketing*, Vol. 33 No. 7, pp. 498-506, doi: 10.1108/JCM-04-2016-1775.

Lee, N. and Cadogan, J. (2013), "Problems with formative and higher-order reflective variables", *Journal of Business Research*, Vol. 66 No. 2, pp. 242-247, doi: 10.1016/j.jbusres.2012.08.004.

Lee, Y.J., Ha, S. and Johnson, Z. (2019), "Antecedents and consequences of flow state in e-commerce", *Journal of Consumer Marketing*, Vol. 36 No. 2, pp. 264-275, doi: 10.1108/JCM-10-2015-1579.

Lee, W., O'Cass, A. and Sok, P. (2017), "Unpacking brand management superiority: examining the interplay of brand management capability, brand orientation and formalization", *European Journal of Marketing*, Vol. 51 No. 1, pp. 177-199, doi: 10.1108/EJM-09-2015-0698.

Lindell, M.K. and Whitney, D.J. (2001), "Accounting for common method variance in cross-sectional research designs", *Journal of Applied Psychology*, Vol. 86 No. 1, pp. 114-121.

Liu, J. (2016), *China Gaming Industry Report*, China Book Publishing House, Beijing.

Lu, T. (2016), "Analysis on the copyright regulation of video game plagiarism", *Electronics Intellectual Property*, Vol. 1 No. 2, pp. 32-39.

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCB

Lu, Y., Zhou, T. and Wang, B. (2009), "Exploring Chinese users' acceptance of instant messaging using the theory of planned behavior, the technology acceptance model, and the flow theory", *Computers in Human Behavior*, Vol. 25 No. 1, pp. 29-39, doi: 10.1016/j.chb.2008.06.002.

Mackay, T., Ewing, M., Newton, F. and Windisch, L. (2009), "The effect of product placement in computer games on brand attitude and recall", *International Journal of Advertising*, Vol. 28 No. 3, pp. 423-438, doi: 10.2501/S0265048709200680.

Molyneux, L., Vasudevan, K. and Gil de Zúñiga, H. (2015), "Gaming social capital: exploring civic value in multiplayer video games", *Journal of Computer-Mediated Communication*, Vol. 20 No. 4, pp. 381-399, doi: 10.1111/jcc4.12123.

Mukherjee, S. and Lau-Gesk, L. (2016), "Retrospective evaluations of playful experiences", *Journal of Consumer Marketing*, Vol. 33 No. 5, pp. 387-395, doi: 10.1108/JCM-03-2016-1740.

Murray, J.Y., Gao, G.Y. and Kotabe, M. (2011), "Market orientation and performance of export ventures: the process through marketing capabilities and competitive advantage", *Journal of the Academy of Marketing Science*, Vol. 39 No. 2, pp. 252-269.

Nah, F.F.H., Eschenbrenner, B. and DeWester, D. (2011), "Enhancing brand equity through flow and telepresence: a comparison of 2D and 3D virtual worlds", *MIS Quarterly*, Vol. 35 No. 3, pp. 731-747, doi: 10.2307/23042806.

Nandi, S., Nandi, M.L. and Khandker, V. (2021), "Impact of perceived interactivity and perceived value on mobile app stickiness: an emerging economy perspective", *Journal of Consumer Marketing*, Vol. 38 No. 6, pp. 721-737, doi: 10.1108/JCM-02-2020-3661.

O'Cass, A. and Carlson, J. (2010), "Examining the effects of website-induced flow in professional sporting team websites", *Internet Research*, Vol. 20 No. 2, pp. 115-134, doi: 10.1108/10662241011032209.

Pelet, J.É., Ettis, S. and Cowart, K. (2017), "Optimal experience of flow enhanced by telepresence: evidence from social media use", *Information & Management*, Vol. 54 No. 1, pp. 115-128, doi: 10.1016/j.im.2016.05.001.

Pham, T.T.L., Huang, H.C., Tseng, F.C., Cheng, T.C.E. and Teng, C.I. (2022), "For whom does flow not enhance online gamer loyalty?", *Industrial Management & Data Systems*, Vol. 122 No. 1, pp. 215-234, doi: 10.1108/IMDS-05-2021-0338.

Qin, H., Patrick Rau, P.L. and Salvendy, G. (2009), "Measuring player immersion in the computer game narrative", *International Journal of Human-Computer Interaction*, Vol. 25 No. 2, pp. 107-133, doi: 10.1080/10447310802546732.

Ringle, C.M., Wende, S. and Becker, J.-M. (2022), "SmartPLS 4", available at: www.smartpls.com

Sarstedt, M., Hair, J.F., Jr, Cheah, J.H., Becker, J.M. and Ringle, C.M. (2019), "How to specify, estimate, and validate higher-order constructs in PLS-SEM", *Australasian Marketing Journal*, Vol. 27 No. 3, pp. 197-211.

Shi, S., Chen, Y. and Chow, W.S. (2016), "Key values driving continued interaction on brand pages in social media: an examination across genders", *Computers in Human Behavior*, Vol. 62, pp. 578-589, doi: 10.1016/j.chb.2016.04.017.

Siekpe, J.S. (2005), "An examination of the multidimensionality of flow construct in a computer-mediated environment", *Journal of Electronic Commerce Research*, Vol. 6 No. 1, p. 31.

Soutter, A.R.B. and Hitchens, M. (2016), "The relationship between character identification and flow state within video games", *Computers in Human Behavior*, Vol. 55, pp. 1030-1038, doi: 10.1016/j.chb.2015.11.012.

Statista (2025), "Games—China", available at: www.statista.com/outlook/amo/media/games/china (accessed 31 January 2025).

Stremersch, S., Gonzalez, J., Valenti, A. and Villanueva, J. (2023), "The value of context-specific studies for marketing", *Journal of the Academy of Marketing Science*, Vol. 51 No. 1, pp. 50-65, doi: 10.1007/s11747-022-00872-9.

Su, Y.S., Chiang, W.L., Lee, C.T.J. and Chang, H.C. (2016), "The effect of flow experience on player loyalty in mobile game application", *Computers in Human Behavior*, Vol. 63, pp. 240-248, doi: 10.1016/j.chb.2016.05.049.

Thomala, L.L. (2023), "Gaming in China—statistics & facts", available at: www.statista.com/topics/4642/gaming-in-china/#topicOverview (accessed 25 December 2023).

Von der Pütten, A.M., Klatt, J., Ten Broeke, S., McCall, R., Krämer, N.C., Wetzel, R., Blum, L., Oppermann, L. and Klatt, J. (2012), "Subjective and behavioral presence measurement and interactivity in the collaborative augmented reality game TimeWarp", *Interacting with Computers*, Vol. 24 No. 4, pp. 317-325, doi: 10.1016/j.intcom.2012.03.004.

Voulgari, I. and Komis, V. (2012), "Antecedents of collaborative learning in massively multiplayer online Games", in: Daradoumis, T., Demetriadis, S. and Xhafa, F. (Eds), *Intelligent Adaptation and Personalization Techniques in Computer-Supported Collaborative Learning*, Springer, Berlin, pp. 269-293, doi: 10.1007/978-3-642-28586-8_12.

Yan, Y., Davison, R.M. and Mo, C. (2013), "Employee creativity formation: the roles of knowledge seeking, knowledge contributing and flow experience in web 2.0 virtual communities", *Computers in Human Behavior*, Vol. 29 No. 5, pp. 1923-1932, doi: 10.1016/j.chb.2013.03.007.

Zaman, M., Anandarajan, M. and Dai, Q. (2010), "Experiencing flow with instant messaging and its facilitating role on creative behaviors", *Computers in Human Behavior*, Vol. 26 No. 5, pp.1009-1018, doi: 10.1016/j.chb.2010.03.001.

Further reading

Carlson, J. and O'Cass, A. (2011), "Managing web site performance taking account of the contingency role of branding in multi-channel retailing", *Journal of Consumer Marketing*, Vol. 28 No. 7, pp. 524-531, doi: 10.1108/07363761111181518.

Corresponding author

Mohammad Mahfuzur Rahman can be contacted at: MMRahman@ship.edu

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: **permissions@emeraldinsight.com**

Downloaded from www.emerald.com/jcm/article-pdf/42/6/795/10142878/jcm-04-2024-6755en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAALcwgbQGCSqGSIb3DQEHBqCBpjCBowIBADCB